

Question

How “symplectic” is the geometry of G_2 -manifolds?

On the symplectic side, we have 2-forms ω satisfying $d\omega = 0$, and $X \lrcorner \omega = 0$ implies $X = 0$, and on the torsion-free G_2 -side, we have 3-forms φ satisfying (among other things) $d\varphi = 0$, and (strong nondegeneracy), $(X_1 \wedge X_2) \lrcorner \varphi = 0$ implies X_1, X_2 are linearly dependent.

Well, the fact $3 > 2$ makes things a lot harder. It means instead of Hamiltonian vector fields induced by functions, we would have to start with 1-forms, take d of them to get 2-forms, and that’s a lot less well-behaved. Moreover, the fact symplectic forms are closed 2-forms means that they can be the curvature of the connection of some complex line bundle (called a prequantum line bundle), so if you’re looking for the total space of some symmetry reduction, you know cohomologically what to look for, and this tells you a lot. this has a lot of the “symplectic geometry” encoded in it.

In particular, this leads to really strong topological restrictions on what kind of things can have the algebraic structure like strong nondegeneracy akin to G_2 -manifolds, like:

Proposition (Eckmann-Whitehead 63, Brown-Gray 67)

The only strongly nondegenerate forms are (almost) symplectic forms, volume forms, (possibly split) associative 3-forms, and (possibly split) Cayley 4-forms.

So if you want to emulate symplectic geometry on G_2 -manifolds, somethings will obviously not carry over straightforwardly. What if we find a symplectic manifold encoding a lot of the information?

Definition

Let (M, φ) be a manifold with closed G_2 -structure. Let LM (free loop space) be the space of smooth maps $S^1 \rightarrow M$, and $\text{Emb}(S^1)$ be the space of embeddings (knots).

Choose periodic coordinate t on our circles so our circles have length 1.

LM is a Frechet manifold, modelled on $\mathcal{C}^\infty(S^1, \mathbb{R}^7)$. Its topology is the given by the inverse limit of Hilbert spaces $H^k(S^1, \mathbb{R}^n)$, which is the completion of $\mathcal{C}^\infty(S^1, \mathbb{R}^n)$ wrt the seminorm

$$\|f\|_k = \int_0^1 \sum_{j=1}^k |f^{(j)}(t)|^2 dt$$

$\text{Emb}(S^1)$ is an open subset.

We can get charts as follows: given a loop $\gamma : S^1 \rightarrow M$, we can pullback the tangent bundle to get a vector bundle $\gamma^*TM \rightarrow S^1$, which is oriented since φ induces an orientation. Given $\varepsilon > 0$ small enough, taking tangent vectors of length $< \varepsilon$, we can identify γ^*TM_ε with an open subset of M via the exponential map, so that $L(\gamma^*TM_\varepsilon)$ can be identified with an open subset of LM . But $\gamma^*TM \cong S^1 \times \mathbb{R}^n$ is trivial since it is an oriented bundle over S^1 , and so induces a map $L(\gamma^*TM_\varepsilon) \rightarrow L\mathbb{R}^n = \mathcal{C}^\infty(S^1, \mathbb{R}^n)$, which is our chart. These are nice enough to give

Proposition (McLau)

These charts define a paracompact smooth manifold structure on LM , and are in fact independent of the metric.

The tangent space at $\gamma \in \text{Emb}(S^1)$ is $\Gamma(S^1, \gamma^*TM)$.

The orientation-preserving diffeomorphisms of $\text{Diff}^+(S^1)$ act on the right on LM and $\text{Emb}(S^1)$ by composition.

Proposition

The quotient $\mathcal{K} := \text{Emb}(S^1)/\text{Diff}^+(S^1)$ is an infinite dimensional smooth manifold, locally modeled by

$\mathcal{C}^\infty(S^1, \mathbb{R}^{n-1})$, and the quotient map $\text{Emb}(S^1) \rightarrow \mathcal{K}$ gives a principal $\text{Diff}^+(S^1)$ -bundle.

idea of proof:

given $\gamma : S^1 \rightarrow M$, let T be the open subset of the normal bundle N to the immersion γ containing the zero section of N , such that there exists an immersion $i : T \rightarrow M$ extending γ (thought of as a function on the zero section). This gives a bundle $p : T \rightarrow S^1$. Let U be the open subset of maps $\gamma : S^1 \rightarrow T$ such that $p \circ \gamma \in \text{Diff}^+(S^1)$. Then one can directly definition chase to get $U/\text{Diff}^+(S^1)$ is diffeomorphic to the space of sections of $p : T \rightarrow S^1$, so that such spaces of sections serve as charts. ■

$\mathcal{K}$ is the moduli space of knots. Its tangent space at $[\gamma] \in \mathcal{K}$ is $\Gamma(S^1, N_{S^1/M})$, where the normal bundle is wrt γ . We can define a symplectic form as follows on it:

Let $ev : \text{Emb}(S^1) \times S^1 \rightarrow M$ be the evaluation map, and $\pi_1 : \text{Emb}(S^1) \times S^1 \rightarrow \text{Emb}(S^1)$ be projection. Then we can get a 2-form by integrating φ along the fibers of π_1 :

$$\omega := \int_{S^1} ev^* \varphi = (\pi_1)_*(ev^* \varphi)$$

which acts on $u, v \in T_\gamma \text{Emb}(S^1)$ by

$$\omega_\gamma(u, v) = \int_{S^1} (u \wedge v) \lrcorner \varphi = \int_0^1 \varphi_{\gamma(t)}(u(t), v(t), \frac{d}{dt} \gamma(t)) dt$$

ω is closed since φ is. However, it is nondegenerate on $\mathcal{K}$: two equivalentish ways to see this. When integrating, $(u \wedge v) \lrcorner \varphi$ restricted to a fibre S^1 is not top degree if u or v is tangent to S^1 , so in directions tangent to the orbits of $\text{Diff}^+(S^1)$ we have degeneracy. Equivalently, we can see that ω is basic with respect to $\text{Diff}^+(S^1)$, i.e. (this is only equivalent if $d\omega = 0$) its hook with any vertical vector field is zero, since $(f \cdot \frac{d\gamma}{dt}) \lrcorner \omega = 0$, as $f \frac{d\gamma}{dt}$ and $\frac{d\gamma}{dt}$ are parallel. But basic means that the degeneracy is really due to the $\text{Diff}^+(S^1)$ symmetry only (this is really the definition of a basic differential form), so if we kill the symmetry we kill the degeneracy.

Thus, we get a symplectic form on $\mathcal{K}$ by passing to the quotient.

However, we have to be careful: infinite dimensions means we can only ask that $X \mapsto X \lrcorner \omega$ be injective, though it also has dense image (Brylinski Corollary 3.5.3).

With similar arguments but drawing more arrows, ω really encodes a lot of the cohomological information that φ does, for example, it is exact if φ is exact, and integral if φ is integral, etc. So it is a good candidate to symplectify (M, φ) .

However, it encodes more than property of M implies property of $\mathcal{K}$, the converse often holds:

Lemma (Localization)

If $d\omega = 0$, then φ is closed. (Similar statements hold for integration along fibers in general)

Proof:

We must show $d\varphi_p(v_1, \dots, v_4) = 0$. wlog $v_1, \dots, v_4$ orthonormal, and choose an embedding $\gamma \in \text{Emb}(S^1)$ with $p \in \gamma(S^1)$ s.t. v_1 is tangential, v_2, v_3, v_4 are normal. choose sections $\widehat{v}_2, \widehat{v}_3, \widehat{v}_4 \in \Gamma(S^1, \gamma^* TM)$ that agree at p with v_2, v_3, v_4 . By multiplying $\widehat{v}_2$ with a sequence of functions on S^1 approaching the Dirac delta at p as $\varepsilon \rightarrow 0$, we obtain sections $(\widetilde{v}_2)_\varepsilon$ such that

$$d\varphi_p(v_1, v_2, v_3, v_4) = \lim_{\varepsilon \rightarrow 0} \int_{S^1} ev^*(d\varphi)((\widetilde{v}_2)_\varepsilon, \widehat{v}_3, \widehat{v}_4) = 0$$

since $d\omega = \int_{S^1} ev^*(d\varphi) = 0$. ■

With the same type of approach, many arguments feel finite dimensional (can be reduced to pointwise linear algebra on M) given sufficient analytic hypotheses that allow you to interchange integrals on fibers with your operations.

With similar arguments, one also obtains results like:

Theorem (Lee-Leung)

Coassociative submanifolds of M correspond bijectively to Lagrangian submanifolds of $\mathcal{K}$, where here a Lagrangian submanifold L is a maximal submanifold of $\mathcal{K}$ where ω vanishes, iff ω vanishes and any vector field that is ω -orthogonal to L is tangent along L .

Something with more annoying analytic hypotheses also works for calibrated submanifolds.

Moreover, we can define Hamiltonian vector fields and Poisson brackets with ω for “nice enough” smooth functions f called supersmooth functions (where their derivatives satisfy some technical hypotheses distributionally).

We can also do the rest of Kaehler geometry:

Proposition

The L^2 -metric $g_{[\gamma]}(u, v) = \int_{\gamma(S^1)} g^\varphi(u, v) \mu_{\gamma(S^1)}$ is well-defined and makes $\text{Emb}(S^1) \rightarrow \mathcal{K}$ a Riemannian submersion, where $\text{Emb}(S^1)$ has the usual L^2 -metric.

At least when $M = \mathbb{R}^7$, there should be a lot we can compute here, for example, in the $M = \mathbb{R}^3$ case, Brylinski computes the full curvature tensor in terms of not very nice integrals.

Proposition

Let $\times$ be the 7-dimensional cross product, defined using φ by raising an index. Define an almost complex structure J as follows. Let $[\gamma] \in \mathcal{K}$, $u \in T_{[\gamma]}\mathcal{K} = \Gamma(S^1, N_{S^1/M})$, and let e_1 be a local unit vector field over TS^1 . Define

$$J(u) = e_1 \times u$$

locally.

By expanding all the definitions, we obtain

$$\omega(u, v) = g(Ju, v)$$

where in our model space $\mathcal{C}^\infty(S^1, \mathbb{R}^6)$, J corresponds to multiplication by i in $\mathbb{R}^6 \cong \mathbb{C}^3$.

In turn,

Theorem (Brylinski, Marsden-Weinstein)

For the analogous J defined with the 3-dimensional dimensional cross product, φ being the volume form, its Nijenhuis tensor vanishes identically. (Pretty sure this is true for our case, but need to check some details). However,

Theorem (Lempert)

In the 3-dimensional cross product case, J is NEVER holomorphically integrable for any open subset of $\mathcal{K}$.

(Just say this, however in the 3-dimensional case with twistor theory, Lempert, based on ideas of Drinfeld and LeBrun, was able to show that such a J is “weakly integrable”, in the sense there is a weakly dense subset of each cotangent space (say, at x) such that each element of that set is the derivative of some holomorphic function defined in a neighbourhood of x).

Proposition

wrt J , the map ι that flips the orientation of a knot in $\text{Emb}(S^1)$ is anti-holomorphic.

Theorem (Hitchin)

In the 3-dimensional cross product case, a geodesic γ in $\text{Emb}(S^1)$ has constant curvature if and only if the tangent space at γ is invariant under J .

Altogether, we can combine to get formal Kaehler geometry:

Proposition

$(\mathcal{K}, J, \omega, g)$ is a formally Kaehler infinite dimensional manifold, where ω is type $(1, 1)$.

Theorem (Brylinski)

In the 3-dimensional case, if M is compact and $\int_M \mu \in \mathbb{Z}$, then ω is integral.

Proof:

$\text{Diff}^+(S^1)$ acts on $S^1 \times \text{Emb}(\Sigma)$ diagonally $f \cdot (x, \gamma) = (f^{-1}(x), \gamma \circ f)$. ev then factors through a smooth mapping $e : (S^1 \times \text{Emb}(\Sigma))/\text{Diff}^+(S^1) \rightarrow M$, where the domain is a smooth manifold that gives an S^1 -bundle over $\mathcal{K}$: $q : (S^1 \times \text{Emb}(\Sigma))/\text{Diff}^+(S^1) \rightarrow \mathcal{K}$. Then, $\omega = \int_q e^* \mu$. As all of these operations in cohomology are defined over the integers, the result follows. ■

Corollary

Thus, by the Kostant-Souriau theorem, there is a (prequantum) complex line bundle L over $\mathcal{K}$ (in the 3-dimensional case) whose curvature is $2\pi i \omega$.

This doesn't immediately work for G_2 -manifolds, since $\int_M \varphi$ doesn't work, though its induced volume form should have something to say here.

However, if this generalizes, this would be very important! The application of Kostant-Souriau really tells us then that $[\omega]$ is c_1 of this line bundle, so that we have a de Rham characteristic class that is cohomologically determined by φ !

Finally, we can consider a Hamiltonian action on $\mathcal{K}$ with a moment map. The group $G =$ volume preserving diffeos of M , whose Lie algebra is divergence-free vector fields, acts on $\text{Emb}(S^1)$ and commutes with the action of $\text{Diff}^+(S^1)$, and preserves ω , so descends to an action by symplectomorphisms on $\mathcal{K}$. Moreover, it admits a moment map in the $n = 3, b^1(M) = 0$ case: a map $\mu : \mathcal{K} \rightarrow \mathfrak{g}^*$ satisfying

$$d\langle \mu, \xi \rangle = \xi^\# \lrcorner \omega, \quad \xi \in \mathfrak{g}$$

$$\{\langle \mu, \xi \rangle, \langle \mu, \eta \rangle\} = \langle \mu, [\xi, \eta] \rangle$$

for ξ, η , whenever their Hamiltonian vector fields are super smooth. If we define μ by

$$\langle \mu(\gamma), \xi \rangle = \int_{\gamma(S^1)} \alpha$$

where α is a 1-form such that $d\alpha$ is dual to $\xi^\#$. (what?)

one can prove this is an equivariant moment map by taking formulas in terms of curls, which should work in the G_2 -case.

Wild Speculation

So $\mathcal{K}$ is formally Kaehler, and admits a moment map, so we want to symplectically reduce it. What will we get?

Time to gesture vaguely. So the volume-preserving diffeos should admit a moment map in the G_2 -case. If one believes that this is the base space of a prequantum line bundle, then this moment map should be gauge-theoretically similar to the "curvature" moment map of the space of connections over a Riemann surface, whose symplectic reduction $\mu^{-1}(0)/G$ is the moduli space of flat connections over a Riemann surface. By the work of gauge theorists in the 80s, we know such a moduli space is finite dimensional. However, the way they proved it was finite dimensional was either lots of analysis, (see Appendix to Wells complex geometry book) or by realizing it was diffeomorphic to an appropriate space of unitary representations of a central extension of the fundamental group of the surface (which is really the moduli space of polystable bundles over the surface), so they already knew what the space was supposed to be

when they reduced, and hence could argue finite dimensionality more directly. We don't know what the reduced space should be: one guess is that if "the moment map is the curvature" holds literally enough, it could be the second fundamental form, and then we have the moduli space of totally geodesic closed 1-submanifolds, but this seems optimistic. However, it does feel like such a moduli-space "should" be finite-dimensional, since all our arguments feel finite-dimensional; it seems like it is "only" the infinite-dimensional symmetries that are complicating things. Alternatively, if it is the base of a prequantum line bundle, the common thing to do in finite dimensions is then Kodaira embed the space (where then the prequantum line bundle is (to a high enough tensor power), $\mathcal{O}(1)$), and then pull everything back. This is a common argument in some kinds of gauge theory when you want to prove you have a finite dimensional algebraic thing.